Andres Diaz Aguilar

adiazaguilar@ethz.ch LinkedIn Google Scholar Website

Education

MSc Mechanical Engineering | **ETH Zürich** 09/2024 – Present

- GPA: 5.21/6.00. Coursework: Perception and Learning for Robotics, Probabilistic AI, Vision Algorithms for Mobile Robotics, Computational Control, Model Predictive Control, Embedded Control Systems, Introduction to ML

BSc Mechanical Engineering, Cum Laude | **Eindhoven University of Technology (TU/e)** 09/2021 – 08/2024

- Top 4% of cohort, with the Honors Academy distinction
- Thesis: *Through Thickness Microstructure Reconstruction of Polycrystals Using the Phase Field Method* (graded 9.5/10). Results presented at the *2024 ICMS Multiscale Mechanics and Artificial Microstructures Workshop*. [Link]

Publications & Presentations

*Equal contribution

L. Guo*, **A. M. Diaz Aguilar***, W. Talbot, T. Tuna, M. Hutter, C. Cadena. *CaRLi-V: Camera-RADAR-LiDAR Point-Wise 3D Velocity Estimation*. IEEE Robotics and Automation Practice (RA-P), 2026. [Paper] [Code]

A. M. Diaz Aguilar, A. Lodge, F. Hoekstra. *Unlocking Electrochemical Battery Modelling: An Open-Source Suite Comparison Methodology*. 11th IFAC Symposium on Advances in Automotive Control, 2025. [Paper]

A. Lodge, **A. M. Diaz Aguilar**, F. Hoekstra, S. Wilkins. *Why is PyBaMM so P(y)opular? A Comparative Analysis of Open-Source Battery Modelling Tools*. PyBaMM Battery Modelling Conference, 2025. (Poster)

Reviewer: IEEE ICRA 2027; IFAC AAC 2025.

Relevant Academic Projects

Robotic Systems Lab, ETH Zürich | ROS2, Python, PyTorch, OpenCV, Open3D 11/2025

CaRLi-V: Camera-RADAR-LiDAR Point-Wise 3D Velocity Estimation

- Developed camera–RADAR–LiDAR fusion framework for point-wise dense 3D velocity estimation, leveraging a newly introduced velocity cube RADAR representation for dense radial velocity encoding.
- Reduced mean velocity error by 31% over state-of-the-art scene flow methods in real-world testing.
- Published in IEEE RA-P, to be presented at *ICRA 2027*. Released as an open-source ROS2 package.

Automatic Control Laboratory, ETH Zürich / Tinamu Labs AG | ROS, C++, Eigen, PCL, FAST-LIO2 07/2026

Semester Project: Robust LiDAR-Inertial Odometry in Challenging Sensing Regimes

- Diagnosed FAST-LIO2 failure modes in field deployments, shown to originate from coupled geometric degeneracy, IMU attitude misestimation, and map corruption issues.
- Incorporated LiDAR intensity-augmented registration, motion-aware correspondence filtering, and probabilistic weighting to suppress noise-dominated LiDAR update directions.
- Eliminated odometry divergence in all observed failure cases. Manuscript in preparation for *EEC 2027*.

TU/e Honors Academy (High Tech Systems track) | ROS2, C++, Python, OpenCV, Open3D 02/2024

Mesh-Based Object Detection & Active Perception, supervised by Prof. Elena Torta

- Developed active perception framework for RGB-D-based object detection of objects with known meshes, using point-cloud segmentation and ICP-based mesh alignment to match the mesh to objects in the environment.
- Used detection confidence to select viewpoints that reveal discriminative geometry and resolve ambiguous detections.

Professional Experience

Robotic Systems Lab, ETH Zürich | Siemens NX 08/2026 – Present

Research Assistant

- Working on robot hardware design and the development of experimental platforms to support the lab's research.

Institute for Dynamic Systems and Control, ETH Zürich | SolidWorks 09/2025 – 01/2026

Research Assistant, PerfusionX Lab, Prof. Christopher Onder's group

- Worked on the design, fabrication, and assembly of a novel ex-vivo perfusion machine tailored to medical research.

TNO, Netherlands Org. for Applied Scientific Research | Python, PyBaMM, MATLAB/Simulink 09/2024 – 02/2025

Battery Modeling and Control Internship

- Developed benchmarking framework for physics-based battery models (PBMs) and performed the first comparative study of leading open-source modeling tools. Presented at *IFAC AAC 2025*, published in IFAC-PapersOnLine.
- Developed aging-aware PBM-based fast-charging controller, limiting lithium plating via anode potential constraints.

Skills

Programming languages: Python, C/C++, MATLAB/Simulink, Julia
Frameworks & Tools: ROS/ROS2, PyTorch, OpenCV, Open3D, PCL, Eigen, cvxpy, scikit-learn, Linux, Git
Languages: English (fluent), French (fluent), Spanish (native), Portuguese (native)